

1. Convex shape

1. Convex sets and functions

$C \subseteq \mathbb{R}^d$ is convex if, for all $x, y \in C$ and $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda)y \in C.$$

On a convex C , $f : C \rightarrow \mathbb{R}$ is convex if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Prove a universal claim with arbitrary x, y, λ . Refute it with one escaping segment or one graph point above its chord. A convex feasible set and a convex objective are separate conditions.

Strict convexity makes the inequality strict for $x \neq y$, $0 < \lambda < 1$. If a minimizer exists, it is unique; strict convexity does not prove existence.

Keep existence, global optimality, and uniqueness as separate questions.

2. First-order certificates

1. Differentiable convex objective

Let f be differentiable on an open convex set C . Then f is convex iff

$$f(y) \geq f(x) + \nabla f(x)^T(y-x) \quad \text{for all } x, y \in C.$$

Thus every tangent hyperplane is a global lower bound. In the unconstrained case, $\nabla f(x^*) = 0$ certifies a global minimizer. A zero gradient alone is not enough without convexity.

2. Nonsmooth and constrained cases

For convex f , g is a subgradient at x if

$$f(y) \geq f(x) + g^T(y-x) \quad \text{for all } y,$$

and $\partial f(x)$ is the set of all such g . Then $0 \in \partial f(x^*)$ certifies global optimality. For $f(x) = |x|$, $\partial f(0) = [-1, 1]$.

If C is convex and f is differentiable and convex, a feasible x^* minimizes f over C iff

$$\nabla f(x^*)^T(y-x^*) \geq 0 \quad \text{for all } y \in C.$$

A boundary optimum need not have zero gradient.

3. Curvature and descent

1. Hessian certificates

A symmetric H is PSD ($H \succcurlyeq 0$) if $z^T H z \geq 0$ for every z ; it is PD ($H \succ 0$) if the inequality is strict for nonzero z .

If f is twice continuously differentiable on an open convex domain,

$$f \text{ is convex iff } \nabla^2 f(x) \succcurlyeq 0 \text{ for all } x.$$

$\nabla^2 f(x) \succ 0$ everywhere is sufficient for strict convexity, not necessary.

For $J(w) = \|Aw - b\|_2^2 + \lambda \|w\|_2^2$, $\lambda > 0$,

$$z^T(A^T A + \lambda I)z = \|Az\|_2^2 + \lambda \|z\|_2^2 > 0.$$

2. L -smoothness and descent lemma

f is L -smooth in the Euclidean norm if

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2.$$

Then

$$f(y) \leq f(x) + \nabla f(x)^T(y-x) + \frac{L}{2}\|y - x\|_2^2.$$

For $y = x - \eta \nabla f(x)$,

$$f(y) \leq f(x) - \eta \left(1 - L\frac{\eta}{2}\right) \|\nabla f(x)\|_2^2.$$

Hence $0 < \eta \leq \frac{1}{L}$ is a simple sufficient non-increase range, with strict decrease when the gradient is nonzero. Smoothness is an upper-curvature bound; it is not strict or strong convexity.

3. Gradient method and a diagnostic quadratic

$$w_{t+1} = w_t - \eta \nabla f(w_t), \quad \eta > 0.$$

This is an update rule, not a convergence theorem. For $f(w) = \frac{1}{2}(w-3)^2$ and $e_t = w_t - 3$,

$$e_{t+1} = (1 - \eta)e_t.$$

- $\eta = \frac{1}{2}$: same side, distance halves;
- $\eta = \frac{3}{2}$: overshoots, distance halves;
- $\eta = 2$: alternates without objective decrease;
- $\eta = \frac{5}{2}$: alternates with growing distance.

4. Asymptotic feasibility

1. Big-O and encoded input size

$f(n) = O(g(n))$ means: there exist $c > 0, n_0$ such that $f(n) \leq cg(n)$ for every $n \geq n_0$. Constants may depend on parameters declared fixed, but not on the scaling variable.

For $T(n, d) = 3nd^2 + 7d^3 + 50$:

- if both n, d scale: $O(nd^2 + d^3)$;
- if d is explicitly fixed: $O(n)$.

After fixing a computation model and encoding, runtime is polynomial if it is at most cN^k , where N is encoded input length and c, k do not depend on the instance.

If integer M is binary encoded, $N \approx \log_2 M$; an M -step loop is exponential in N because $M \approx 2^N$. Scanning N values is $O(N)$; enumerating $\{0, 1\}^d$ costs 2^d evaluations. Faster hardware changes constants, not the complexity class.

2. Gap and three-axis audit

A computational–statistical gap occurs when information or sample guarantees support the target accuracy, while no efficient method is available under the same assumptions.

One slow implementation does not prove inherent hardness.

- **Statistical**: do the data assumptions support an accuracy statement?
- **Optimization**: do shape and curvature license the optimum or step?
- **Computational**: is the implementation polynomial in every encoded input?

Do not infer runtime from Hoeffding, data sufficiency from convexity, or statistical accuracy from a polynomial-time algorithm.